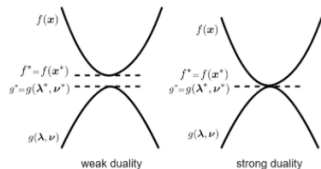
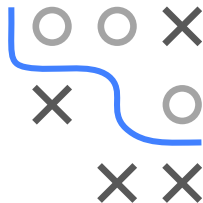


Optimization in Machine Learning

Constrained Optimization

Nonlinear programs and Lagrangian



Learning goals

- Lagrangian for general constrained optimization
- Geometric intuition for Lagrangian duality
- Properties and examples

NONLINEAR CONSTRAINED OPTIMIZATION

- Previous lecture: **Linear programs**

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) := \mathbf{c}^T \mathbf{x} \quad \text{s.t. } \mathbf{Ax} \leq \mathbf{b}, \mathbf{Gx} = \mathbf{h}$$

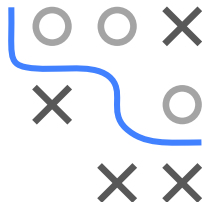
- Related to its (Lagrange) dual formulation by the *Lagrangian*

$$\mathcal{L}(\mathbf{x}, \alpha, \beta) = \mathbf{c}^T \mathbf{x} + \alpha^T (\mathbf{Ax} - \mathbf{b}) + \beta^T (\mathbf{Gx} - \mathbf{h})$$

- **Weak duality:** For $\alpha \geq 0$ and β :

$$f(\mathbf{x}^*) \geq \min_{\mathbf{x} \in \mathcal{S}} \mathcal{L}(\mathbf{x}, \alpha, \beta) \geq \min_{\mathbf{x} \in \mathbb{R}^d} \mathcal{L}(\mathbf{x}, \alpha, \beta) =: g(\alpha, \beta)$$

- **Recall:** Implicit domain constraint in *Lagrange dual function*
 $g(\alpha, \beta)$



GENERAL CONSTRAINED OPTIMIZATION

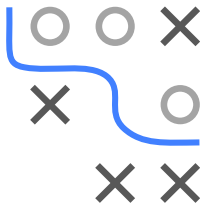
- General form of a constrained optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, k, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, \ell$$

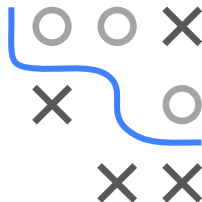
- Functions f, g_i, h_j generally nonlinear
- Transfer the Lagrangian from linear programs:

$$\mathcal{L}(\mathbf{x}, \alpha, \beta) := f(\mathbf{x}) + \sum_{i=1}^k \alpha_i g_i(\mathbf{x}) + \sum_{j=1}^{\ell} \beta_j h_j(\mathbf{x})$$

- Dual variables $\alpha_i \geq 0$ and β_i are also called *Lagrange multipliers*



CONSTRAINED PROBLEMS: THE DIRECT WAY



- Simple constrained problems can be solved in a direct way

Example 1:

$$\min_{x \in \mathbb{R}} 2 - x^2 \quad \text{s.t. } x - 1 = 0$$

- **Solution:** Resolve the constraint by $x - 1 = 0 \Rightarrow x = 1$
and insert it into the objective: $x^* = 1, f(x^*) = 1$

CONSTRAINED PROBLEMS: THE DIRECT WAY

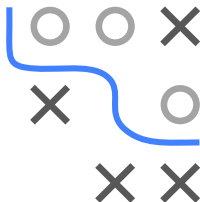
Example 2:

$$\min_{\mathbf{x} \in \mathbb{R}^2} -2 + x_1^2 + 2x_2^2 \quad \text{s.t.} \quad x_1^2 + x_2^2 - 1 = 0$$

- **Solution:** Resolve the constraint $x_1^2 = 1 - x_2^2$ and insert it into the objective

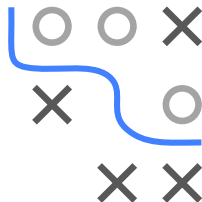
$$f(x_1, x_2) = -2 + (1 - x_2^2) + 2x_2^2 = -1 + x_2^2$$

- $\Rightarrow$ Minimum at $\mathbf{x}^* = (\pm 1, 0)^T$
- However, direct way mostly not possible



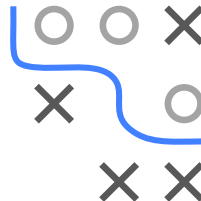
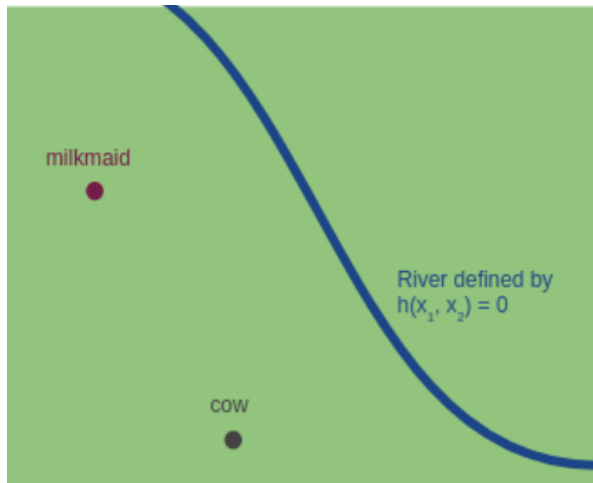
A CLASSIC EXAMPLE: MILKMAID PROBLEM

- **Question 1:** Is there a general recipe for solving general constrained nonlinear optimization problems?
 - **Question 2:** Can we understand this recipe geometrically?
 - **Question 3:** How does this relate to the Lagrange function approach?
-
- For this purpose, we consider the classical “milkmaid problem”
 - Assume a milkmaid is sent to the field to get the day’s milk
 - The milkmaid wants to finish her job as quickly as possible
 - However, she has to clean her bucket first at the nearby river



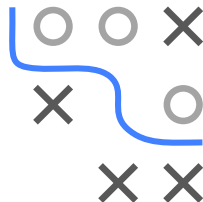
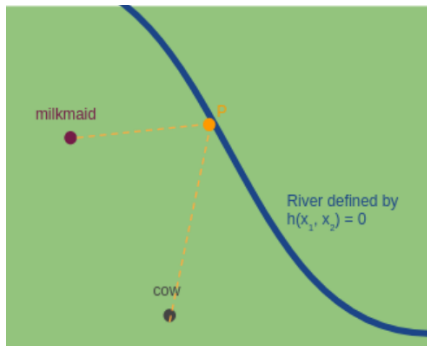
MILKMAID PROBLEM

- Where is the best point P to clean her bucket?



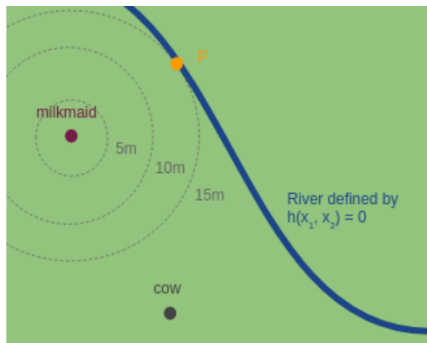
MILKMAID PROBLEM

- **Aim:** Find point P at the river for minimum total distance $f(P)$
- $f(P) := d(M, P) + d(P, C)$ (d measures distance)
- $h(P) = 0$ describes the river

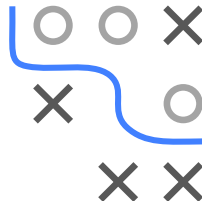


MILKMAID PROBLEM

- **Question:** How far can the milkmaid get for a fixed total distance $f(P)$?
- **Assume:** We only care about $d(M, P)$



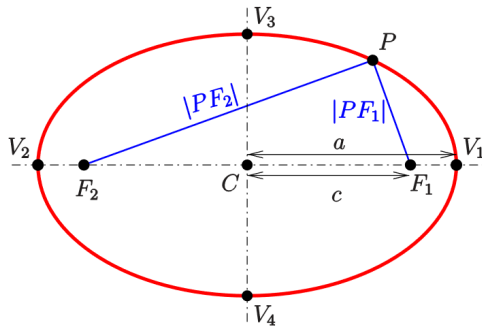
- **Observe:** Radius of circle touching river first is the minimal distance



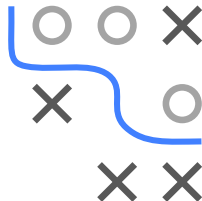
MILKMAID PROBLEM

- For $f(P) = d(M, P) + d(P, C)$: Use an **ellipse**
- **Def.:** Given two focal points F_1, F_2 and distance $2a$:

$$E = \{P \in \mathbb{R}^2 \mid d(F_1, P) + d(P, F_2) = 2a\}$$

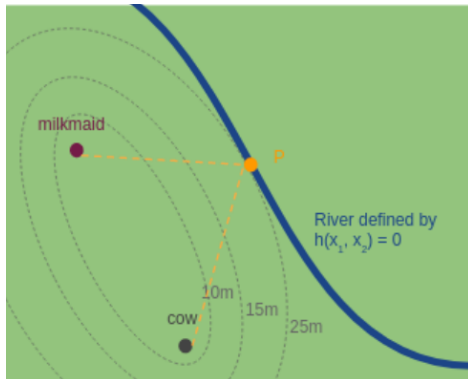


$$|PF_1| + |PF_2| = 2a$$

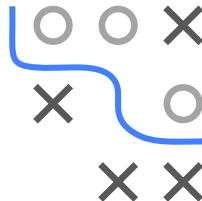


MILKMAID PROBLEM

- Let M and C be focal points of a collection of ellipses
- Find **smallest** ellipse touching the river $h(x_1, x_2)$
- Define P as the touching point



- **Question:** How can we make sense of this mathematically?

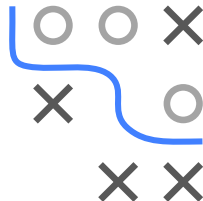
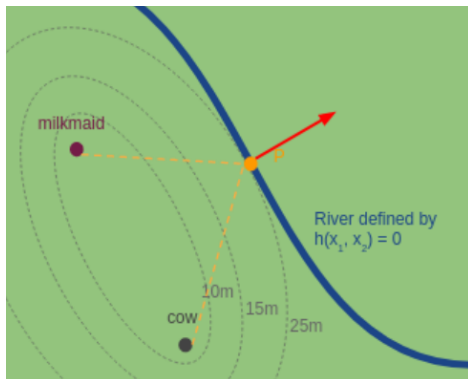


MILKMAID PROBLEM

- **Recall:** Gradient is normal (perpendicular) to contour lines
- Since contour lines of f and h touch, gradients are parallel:

$$\nabla f(P) = \beta \nabla h(P)$$

- Multiplier β is called **Lagrange multiplier**



LAGRANGE FUNCTION

- **General:** Solve problem with single equality constraint by:

$$\nabla f(\mathbf{x}) = \beta \nabla h(\mathbf{x}) \quad \text{and} \quad h(\mathbf{x}) = 0$$

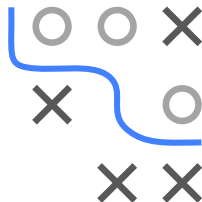
- **First line:** Parallel gradients | **Second line:** Constraint
- **Observe:** Above system is equivalent to

$$\nabla \mathcal{L}(\mathbf{x}, \beta) = \mathbf{0}$$

for **Lagrange function** (or **Lagrangian**) $\mathcal{L}(\mathbf{x}, \beta) := f(\mathbf{x}) + \beta h(\mathbf{x})$

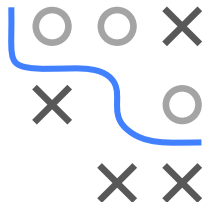
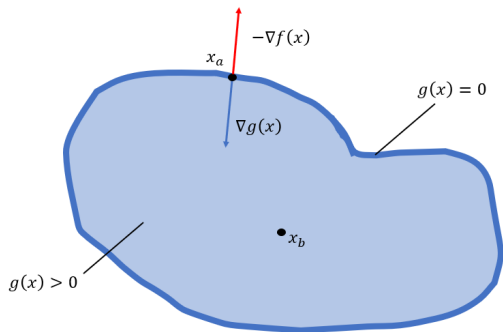
- Indeed:

$$\begin{pmatrix} \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \beta) \\ \nabla_{\beta} \mathcal{L}(\mathbf{x}, \beta) \end{pmatrix} = \begin{pmatrix} \nabla f(\mathbf{x}) + \beta \nabla h(\mathbf{x}) \\ h(\mathbf{x}) \end{pmatrix}$$



LAGRANGE FUNCTION

- Idea can be extended to **inequality** constraints $g(\mathbf{x}) \leq 0$
- There are two possible cases for a solution:
 - Solution $\mathbf{x}_b$ inside feasible set: constraint is inactive ($\alpha_b = 0$)
 - Solution $\mathbf{x}_a$ on boundary $g(\mathbf{x}) = 0$: $\nabla f(\mathbf{x}_a) = \alpha_a \nabla g(\mathbf{x}_a)$ ($\alpha_a > 0$)



LAGRANGE FUNCTION AND PRIMAL PROBLEM

- General constrained optimization problems:

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t. } g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, k, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, \ell$$

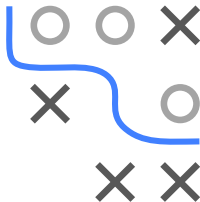
- Extend Lagrangian ($\alpha_i \geq 0$, β_j Lagrange multipliers):

$$\mathcal{L}(\mathbf{x}, \alpha, \beta) := f(\mathbf{x}) + \sum_{i=1}^k \alpha_i g_i(\mathbf{x}) + \sum_{j=1}^{\ell} \beta_j h_j(\mathbf{x})$$

- **Equivalent** primal problem:

$$\min_{\mathbf{x}} \max_{\alpha \geq 0, \beta} \mathcal{L}(\mathbf{x}, \alpha, \beta)$$

- **Question:** Why?



LAGRANGE FUNCTION AND PRIMAL PROBLEM

- For simplicity: Consider only single inequality constraint $g(\mathbf{x}) \leq 0$
- If $\mathbf{x}$ **breaks** inequality constraint ($g(\mathbf{x}) > 0$):

$$\max_{\alpha \geq 0} \mathcal{L}(\mathbf{x}, \alpha) = \max_{\alpha \geq 0} f(\mathbf{x}) + \alpha g(\mathbf{x}) = \infty$$

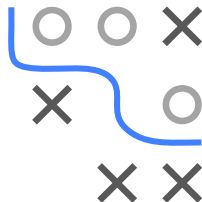
- If $\mathbf{x}$ **satisfies** inequality constraint ($g(\mathbf{x}) \leq 0$):

$$\max_{\alpha \geq 0} \mathcal{L}(\mathbf{x}, \alpha) = \max_{\alpha \geq 0} f(\mathbf{x}) + \alpha g(\mathbf{x}) = f(\mathbf{x})$$

- Combining yields **original formulation**:

$$\min_{\mathbf{x}} \max_{\alpha \geq 0} \mathcal{L}(\mathbf{x}, \alpha) = \begin{cases} \infty & \text{if } g(\mathbf{x}) > 0 \\ \min_{\mathbf{x}} f(\mathbf{x}) & \text{if } g(\mathbf{x}) \leq 0 \end{cases}$$

- Similar argument holds for equality constraints $h_i(\mathbf{x})$



EXAMPLE: LAGRANGE FUNCTION FOR QPS

- We consider quadratic programming

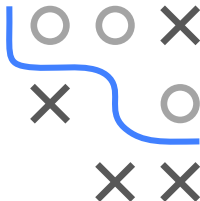
$$\min_{\mathbf{x}} f(\mathbf{x}) := \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} \quad \text{s.t. } h(\mathbf{x}) := \mathbf{C} \mathbf{x} - \mathbf{d} = \mathbf{0}$$

with $\mathbf{Q} \in \mathbb{R}^{d \times d}$ symmetric, $\mathbf{C} \in \mathbb{R}^{\ell \times d}$, and $\mathbf{d} \in \mathbb{R}^{\ell}$

- Lagrange function: $\mathcal{L}(\mathbf{x}, \beta) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \beta^T (\mathbf{C} \mathbf{x} - \mathbf{d})$
- Solve

$$\nabla \mathcal{L}(\mathbf{x}, \beta) = \begin{pmatrix} \partial \mathcal{L} / \partial \mathbf{x} \\ \partial \mathcal{L} / \partial \beta \end{pmatrix} = \begin{pmatrix} \mathbf{Q}\mathbf{x} + \mathbf{C}^T \beta \\ \mathbf{C}\mathbf{x} - \mathbf{d} \end{pmatrix} = \mathbf{0} \quad \Leftrightarrow \quad \begin{pmatrix} \mathbf{Q} & \mathbf{C}^T \\ \mathbf{C} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \beta \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{d} \end{pmatrix}$$

- **Observe:** Solve QP by solving a linear system



LAGRANGE DUALITY

- **Dual problem:**

$$\max_{\alpha \geq 0, \beta} \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \alpha, \beta)$$

- Define **Lagrange dual function** $g(\alpha, \beta) := \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \alpha, \beta)$
 - Important characteristics of the dual problem:
 - **Convexity** (pointwise minimum of *affine* functions)
 - Gives methods based on dual solutions
 - Might be computationally inefficient (expensive minimizations)
 - **Weak duality:** $f(\mathbf{x}^*) \geq g(\alpha^*, \beta^*)$
 - **Strong duality** if primal problem satisfies *Slater's condition*⁽¹⁾:
 $f(\mathbf{x}^*) = g(\alpha^*, \beta^*)$
- (1) **Slater's condition:** Primal problem convex and “strictly feasible”
($\exists \mathbf{x} \forall i : g_i(\mathbf{x}) < 0$)

